



**Verified Carbon
Standard**

VALIDATION REPORT

“NKHOTAKOTA SOLAR PV PROJECT”



Report ID	EPL-VCS-2023-0031
Project title	Nkhotakota Solar PV project
Project ID	4486
Crediting period	29-July-2023 to 28-July-2030
Original date of issue	01-August 2024
Most recent date of issue	07-February 2025
Version	2.0
VCS Standard Version	4.7
Client	South Pole Carbon Asset Management Ltd.
Prepared by	EcoLance Private Limited

Approved by	 Rekha Menon- Director
Work carried out by	Team Leader/Technical Expert/Validator: Champok Buragohain Team member: Prem Kumar Pokhrel

Summary:

EcoLance Private Limited (EPL), appointed by South Pole Carbon Asset Management Ltd., has performed the validation of the greenhouse gas emission reductions reported for the project activity 'Nkhotakota Solar PV project' VCS 4486, with regard to the relevant requirements for VCS activities.

The project activity is a grid connected solar PV project which replaces equivalent amount of electricity otherwise would have generated in fossil fuel dominated grid. Therefore, the project reduces greenhouse gas emissions from fossil fuel-based power plants connected to grid.

The objective of the validation is to have an independent review of ex-post determination of the monitored reductions in GHG emission. Validation was conducted using EPL's procedures in line with the requirements specified in the VCS program guide version 4.4, VCS standard Version 4.7, VCS validation & verification manual version 3.2, ISO14064-3: 2019, CDM M&P, the latest version of the CDM validation & verification standard, and relevant discussion of the COP/MOP and the CDM EB and applying standard auditing techniques. The validation consists of desk review, on-site assessment and the resolution of outstanding issues and the issuance of the final validation report and certification.

During this validation, 08 Corrective Action Request (CARs) and 02 Clarification Requests (CLs) were identified related to operation, GHG emission reduction estimation, monitoring and baseline scenario of the VCS project activity in relation to all relevant VCS requirements for the project activity and the applied baseline and monitoring methodology. These CARs and CLs are successfully closed after necessary corrections/clarifications by the client. The same has been discussed in Appendix B of this validation report.

VVB found no uncertainties during this validation of the project activity.

In conclusion, it is EPL's opinion that the project activity 'Nkhotakota Solar PV project', VCS project ID 4486, meets all relevant requirements for VCS standard, v4.7 and related guidelines and correctly applies the baseline and monitoring methodology ACM0002 'Grid-connected electricity generation from renewable sources', Version Number 21. The project is expected to result in GHG emission reductions of average 22,148 tCO₂e per year in the first 7 years of crediting period.

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1 INTRODUCTION

1.1 Objective

South Pole Carbon Asset Management Ltd. has appointed EcoLance Private Limited to carry out validation of the project “Nkhotakota Solar PV project” VCS reference number 4486. This report summarizes the findings of the validation of the project, performed on the basis of VCS Requirements and UNFCCC criteria for CDM, as well as criteria given to provide for consistent project operations, monitoring and reporting.

The objective of validation is to have an independent evaluation of a project activity by an accredited validation and verification body against the requirements of the VCS Version 4.7 and GHG program applied, on the basis of the project design document. In particular, the project’s baseline, monitoring plan, and the project’s compliance with relevant VCS standard Version 4.7, VCS program guide version 4.4, GHG program requirements and host Party criteria are validated in order to confirm that the project design, as documented, is sound and reasonable and meets the identified criteria. Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of Verified Carbon Units (VCUs).

1.2 Scope and Criteria

The validation scope is to review the VCS PD against VCS Version 4.7 Requirements and relevant UNFCCC criteria. EPL, based on VCS validation and verification manual and latest CDM validation and verification standard, VCS standard 4.7, EPL has employed a rule-based approach (as criteria) in the validation, focusing on the identification of significant reporting rules and the reliability of the project monitoring.

Validation is not meant to provide any consultancy towards the project participants. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the monitoring.

1.3 Reasonableness of Assumptions

In order to ensure a complete, transparent and timely execution of the validation task the team leader has planned the complete sequence of events necessary to arrive at a substantiated final validation opinion. Various tools have been established in order to ensure an effective validation planning.

Materiality Threshold:

The validation assessment was conducted to provide a reasonable level of assurance of compliance against the defined audit criteria and materiality threshold for validation as defined in the para 3.10.2 and 4.1.10 (4) of VCS standard /04/ as below:

	Threshold	Related to
☐	1 %	Emission reductions or removals for large projects achieving a total emission reduction or removal of greater than 300,000 tonnes of carbon dioxide equivalent per year
☒	5 %	Emission reductions or removals for projects achieving a total emission reduction or removal of 300,000 tonnes of carbon dioxide equivalent per year or less.

Risk Analysis and validation planning:

Before starting the on-site assessment, the team leader has assessed the nature, scale and complexity of the validation tasks by carrying out a strategic analysis of all activities relevant to the project activity. The team leader has collected and reviewed the information relevant to the project activities and assessed the team competency to carry out the validation and ensured that it is able to conduct the necessary risk analysis. Based on the documents review and the stage of the project, potential risks that are material for the GHG emission estimation were identified and potential mitigation measures were identified before the field validation starts. Potential reporting risks and necessary auditing procedures for residual risk areas identified and assessed are tabulated below:

No	Risk that could lead to material errors, omissions or misstatements	Assessment of the potential risk		Assessment of the records/information/interview with personnel to check controls/ mitigation measures
		Risk level	Justification	
1	Human Error: Recording and reporting of the information in the ER spreadsheet.	Medium	Although the input values in ER spreadsheet are taken from approved DPR and third party reports, there can be human error while transferring the data into excel sheet as	The representative of PP and project operators are interviewed on ER calculation approach and assumptions and the data management approach/systems and QA/QC to manage and transfer data in the database and utilization of it in ER calculation.

			part of ex-ante estimation of emission reduction worksheet. Although, experience consultant has been appointed to minimize the risk. Hence, the risk level is medium.	
2	Information System: Use of spread sheets without adequate controls related to data changes/ updates, version tracking, traceability, security	Medium	Traceability is expected to be maintained although existence of a system is required to be checked.	The audit team has interviewed the team to check the accuracy of system and QA/QC in place.

Validation team formation:

The appropriate validation team was formed with qualified members, qualified in accordance with EPL's qualification procedure for assessing the project for crediting period renewal considering the potential risks and potential mitigation measures for those risks. The validation team and the technical reviewer consist of the following personnel:

Role/Qualification	Last Name	First Name
Team Leader/Technical Expert/Validator	Buragohain	Champok
Team member	Pokhrel	Prem Kumar
Technical Reviewer	Murty	Eswar

Consideration of materiality and the reasonableness in conducting validation:

Based on the validation planning, the audit has been carried out. This is performed by conducting on site assessment and obtaining documentary evidences and other information and explanations to achieve reasonable assurance on emission reduction estimation ex-ante

and the reasonableness of the assumed parameter of interest in line with the applied methodology and supporting evidences. From the audit findings, it is found that the GHG emission reduction estimation ex-ante is fairly stated and are in compliance with the standards and guidelines of VCS /03/, /04/, /05/ and the applied methodology and associated tools /07/. The assessment was done through relevant document review as given in section 2.2 of this report and the corrective actions/clarifications as given in appendix B of this report. The final validation report before being submitted to the client was subjected to an independent internal technical review to confirm that all validation activities had been completed according to the pertinent EPL's instructions, with a reasonable level of assurance. The technical review was performed by a technical reviewer qualified in accordance with EPL's qualification procedure. The emission reduction estimation shown ex-ante and the parameters used for estimation are validated through the applied methodology and other VCS rules and requirement. Therefore, there is no material changes in reporting of ER calculation in final version (version 2.0) of the PD /01/ by PP. So, the audit was done appropriately and sufficiently and the information and the parameter used for the estimates the GHG emission reduction with the outcome of future activities is free of material errors, omissions, misrepresentations and are of reasonable as claimed by PP.

1.4 Summary Description of the Project

This project is a 26 MWp (21 MWac) grid-connected solar PV plant. The project uses photovoltaic solar power technology (PV). The associated infrastructure including the sub-station and overhead power lines connected to the Malawian ESCOM's 66 kV transmission line with the aim to bolster the Malawian grid's access to clean and renewable energy. The project activity is an initiative of Serengeti Nkhotakota Limited located approximately 4km via road to the northwest of Nkhotakota, Malawi just off the M18 road between Nkhotakota and Kasungu. The geo-coordinates of the project location is 12°57'50.4"S 34°15'19.9"E.

In the absence of the project activity the equivalent amount of electricity generated from the project activity would have been generated by the Malawian grid connected power plants encompassing fossil fuel-based power plants and stand by diesel power plants.

The starting date of the project activity is 29/07/2023. The start date represents the date of commissioning of the project resulting the day from which the projects start removing GHG emissions. The start date is correctly considered as per VCS standard paragraph 3.8 /04/ and the audit team confirm the date to be corrected after verifying the commissioning certificate /16/ and generation electricity records /35/.

The project adopts the renewable crediting period of 7 years, twice renewable for total of 21 years. The expected average annual emission reductions are 22,148 tCO₂e per annum over the first 7 years of crediting period. Accordingly, the project is expected to generate 155,036 tCO₂e emission reduction during its first 7 years of crediting period.

2 VALIDATION PROCESS

2.1 Method and Criteria

The Validation was conducted using EPL's procedures, in line with the requirements of VCS Version 4.7 and the requirements specified in the CDM M&P, the latest version of the CDM Validation and Verification Standard, VCS validation and verification manual version 3.2 and relevant decisions of the COP/MOP and the CDM EB and applying standard auditing techniques.

The scope of the activity is to 'validate the project to determine conformance with the VCS Program rules and evaluate the reasonableness of assumptions, limitations and methods that support a statement about the outcome of future activities.

For this, the validation consisted of the following major three phases:

- Legal contracting with PP, Document review and planning;
 - Contract with PP for the scope of the activity
 - Desk review of the PD /01/, ER estimation sheet /02/, VERRA registry communications agreement /36/, UNFCCC website /08/, Gold Standard registry /09/, Verra registry /10/.
 - Assessment of materiality threshold based on VCS standard, version 4.5 /04/, risk analysis, audit planning including the on-site planning and validation team appointment.
- On-site assessment and supporting documents review
 - Onsite assessment of project technology to identify operational status, specification of technology implemented as indicated in PD /01/, the baseline and other supporting parameters and cross-checking the data recording system.
 - Interview with PP and project operators regarding the project implementation, GHG accounting systems, monitoring, project management, repair & maintenance, data management, ownership of the activity, double counting issues, QA/QC and calibration of measuring equipment etc.

- Resolution of any material discrepancy and the issuance of the final validation report for renewable of crediting period.
 - Issuance of the findings (CARs/CLs) based on desk study and on-site assessment regarding any discrepancy between PD and the actual implementation situation.
 - Prepare final validation report based on the assessment, resolution of findings and technical reviews.

The following sections outline each step in more detail. The major milestones for the validation are provided below:

- Legal contract with PP for the scope of activity: 31st May 2023 (final signed)
- First version of Project Description and Emission Reduction estimation received: 06 June 2023
- Submission of Notice of Validation/Verification Service (NOVS): 18th October 2023
- Audit Plan communication to PP: 07 November 2023
- On-site Assessment: 15 November 2023
- Issuance of draft Validation Report: 30 November 2023
- Issuance of First Version of Final Validation Report: 01 August 2024

2.2 Document Review

The VCS PD version 2.0 of 05/04/2024, version 3.0 of 05/07/2024, version 4 of 06/02/2025 /01/, in particular the applicability of the methodology, the baseline determination, the additionality of the project activity, the starting date of the project, the monitoring plan, the emission reduction calculations provided in the form of a spreadsheet /02/ were assessed as part of the validation.

Sl. No.	Document's Name
/01/	SouthPole Carbon: VCS PD for the project "Nkhotakota Solar PV project" version 2.0 of 20-March-2024, version 3.0 of 18 July 2024, version 06 February 2025
/02/	SouthPole Carbon: Emission Reduction Calculation Spreadsheet (ERSheet_Nkhotakota_20240117_V1.0.xlsx)
/03/	VCS: VCS Program Guide, version 4.4 of 29-August-2023
/04/	VCS: VCS Standard, version 4.7
/05/	VCS: Project Description Template, version 4.3
/06/	VCS: VCS Guidance-Validation and Verification Manual, Version 3.2
/07/	CDM Executive Board: ACM0002 'Grid-connected electricity generation from renewable sources' version 21

/08/	UNFCCC website for project Activity: https://cdm.unfccc.int/Projects/projsearch.html
/09/	Gold Standard Registry: https://registry.goldstandard.org/projects?q=&page=1
/10/	VERRA Registry: https://registry.VERRA.org
/11/	VCS: Validation Report Template, version 4.4: https://verra.org/documents/vcs-validation-report-template-v4-4/
/12/	CDM TOOL 01: Tool for demonstration and assessment of additionality, version 7
/13/	UNFCCC: CDM TOOL 07: Tool to calculate the emission factor for an electricity system, version 7
/14/	UNFCCC: CDM TOOL 24: Common Practice
/15/	UNFCCC: CDM TOOL 27: Investment Analysis, version 7
/16/	Electricity Supply Corporation of Malawi Limited: Commissioning Certificate of 21 MW AC solar power plant by Serengeti Nkhotakota Limited (formerly Phanes Energy Renewables Nkhotakota Limited) dated 20 September 2023
/17/	Electricity Supply Corporation of Malawi Limited & Serengeti Nkhotakota Limited (formerly Phanes Energy Renewables Nkhotakota Limited): Power Purchase agreement (PPA) for sale/purchase of electricity from 21 MWac solar pv plant at Nkhotakota, Malawi dated 14 th February 2019
/18/	Global Carbon Council: Project Portal (https://projects.globalcarboncouncil.com)_
/19/	Suntech: Technical Specification Document of 445 Wp monocrystalline solar module
/20/	Overseas Private Investment Corporation: Loan commitment letter to Serengeti Nkhotakota Limited for 21 MW solar PV plant located in Nkhotakota, Malawi dated 20 December 2018
/21/	Registrar of Companies, Govt. of Malawi: Certificate of Incorporation of 'Serengeti Nkhotakota Limited' dated 02 May 2018
/22/	RINA Tech Africa (Pty) Ltd: Third party completion certificate of the 21 MWac solar plant by Serengeti Nkhotakota Limited (formerly Phanes Energy Renewables Nkhotakota Limited) as per PPA, dated 06 September 2023.
/23/	Energy Regulation Act of Malawi: https://malawilii.org/akn/mw/act/2004/20/eng@2014-12-31

/24/	Electricity Act Malawi: https://malawilii.org/akn/mw/act/2004/22/eng@2017-12-31#:~:text=(4)Except%20with%20the%20permission,or%20distribution%20line%20has%20been
/25/	IPP Framework of Malawi: https://rise.esmap.org/data/files/library/malawi/Renewable%20Energy/IPP%20Framework.pdf
/26/	Serengeti Nkhotokia and South Pole Carbon Asset Management Ltd.: Emission reduction purchase agreement
/27/	Lease Rent Agreement
/28/	Stakeholder Consultation document & Proof
/29/	Serengeti Nkhotakota Limited: IRR worksheet prepared for the project activity (4.4Updated_IRR_Nkhotakota_20240705)
/30/	Mott Macdonald: Due diligence report for 21 MWac Solar PV plant in Nkhotakota, Malawi dated January 2019
/31/	IPCC: IPCC 2006 guidelines for GHG inventory https://www.ipcc-nggip.iges.or.jp/public/2006gl/
/32/	Project link in VERRA: https://registry.verra.org/app/projectDetail/VCS/4486
/33/	Serengeti Nkhotakota Limited: Declaration that the project is not rejected by any other GHG programme or listed in any other GHG programme, dated 17 January 2024
/34/	Government of Malawi: Environmental Impact Assessment Requirements
/35/	Serengeti Nkhotakota Limited: Electricity generation record for the month of July and August 2023
/36/	VERRA REGISTRY COMMUNICATIONS AGREEMENT by SERENGETI NKHOTAKOTA LTD. & SOUTH POLE CARBON ASSET MANAGEMENT LTD. dated 10 May 2023.
/38/	UNFCCC: CDM Validation and verification standard for project activities, version 3
/39/	UN Trade and Development: UN list of least developed countries https://unctad.org/topic/least-developed-countries/list
/40/	UNFCCC: Harmonized IFI Default Grid Factors 2021 v3.2
/41/	Director of occupational safety, health and welfare: Certificate of registration of workplace occupational safety, health and welfare, dated 14 February 2023

/42/	Regional Commissioner for Land: Land rent receipt for the period from July 2022 to March 2023
/43/	International Monetary Fund (IMF): Inflation, average consumer prices for Malawi: https://www.imf.org/en/Publications/WEO/weo-database/2021/October/weo-report?c=676&s=PCPI,PCPIPCH,PCPIE,PCPIEPCH.&sy=2023&ey=2026&ssm=0&scsm=1&sc=0&ssd=1&ssc=0&sic=0&sort=country&ds=.&br=11
/44/	Wood Group UK Limited: Final Report Nkhotakota Solar PV Project- Environmental and Social Impact Assessment, dated 29 June 2019
/45/	Serengeti Energy: https://serengeti.energy
/46/	Serengeti Nkhotakota Limited: Employment records
/47/	Serengeti Nkhotakota Limited: Audited balance sheet for the project
/48/	National Water Resource Authority: Licence for groundwater abstraction dated 16/12/2022
/49/	Serengeti Nkhotakota Limited & Sunel Limited: Operation and Maintenance agreement for Nkhotakota project dated 26 May 2020
/50/	PWC: https://taxsummaries.pwc.com/malawi/corporate/taxes-on-corporate-income
/51/	Engineering Procurement & Construction contract between Serengeti Nkhotakota Limited, Sunel Limited & Sunil UK Limited dated 14 April 2020
/52/	International Renewable Energy Agency: Energy Profile: Malawi https://www.irena.org/-/media/Files/IRENA/Agency/Statistics/Statistical_Profiles/Africa/Malawi_Africa_RE_SP
/53/	VERRA: https://registry.verra.org/app/search/VCS
/54/	UNFCCC: AHSA-001 'IFI Approach to GHG accounting for renewable energy projects, version 2.0

2.3 Interviews

During the validation, VVB followed physical inspection, documents check and interview with PP and the project operators to check if the reported information in the PD /01/ are correct and followed due process of ER calculation in line with the applied methodology and associated tools /07/ and other VCS requirements /03/, /04/, /05/, /06/. VVB has checked whether the monitoring systems are in place as indicated in the updated PD /01/ and the systems installed in the previous monitoring period are still valid. Other parameters such as

the baseline practices and any repair/maintenance activities, grievance redress systems, mechanism and the ongoing stakeholder consultation are in place.

Further, VVB conducted the interview to cross-check the project implementation, management and compliance with project document /01/. The table below list the detail of interview taken with the PP and the individuals involved in preparing PD, management systems and monitoring systems.

	Date	Name, Role	Organization	Scope of interview
/a/	15 November 2023	Paul Mhanw Site In-charge	Serengeti Energy	Project Ownership, Major milestones, project financing, implementation and operation, roles and responsibilities etc. Grievance handling mechanism, legal compliance, environmental impacts. Government rules/requirements on environmental assessments and others
/b/	15 November 2023	Raghav Das - Consultant	South Pole Carbon	Project documents and ER estimation.
/c/	15 November 2023	Julius Kamau- Consultant	South Pole Carbon	Applicability of methodology and baseline, VCS rules and applicability. Double counting issues.
/d/	15 November 2023	Christine Banda- Dispatch Engineer	Serengeti Energy	Monitoring and recording of data

/e/	15 November 2023	Emmanuel Lison-Maintenance Engineer	Serengeti Energy	parameters, QA/QC procedures, calibration of energy meters, maintenance and operation.
/f/	15 November 2023	Jusice Nkhoma	Community Member	Stakeholder engagement, impact on local recourses, Grievance handling mechanism, social & environmental impacts
/g/	15 November 2023	Loviness Malenga	Community Member	
/h/	15 November 2023	Farness Nkhoma	Community Member	

2.4 Site Visits

To assess the information included in the project documentation and to gain additional information regarding the compliance of the project with the relevant criteria applicable for validation of the project, EPL conducted the site visit at project site and PP's office on 15th November 2023. During site visit, validation team has interviewed project proponent, assessed the project design and implementation and carried out interview with PP and project operators and personals involved in preparing the documents assessed following information:

- Consistency between the information as contained in project document /01/ and ex-ante ER calculation sheet /02/.
- Check if the project design, construction, operation and monitoring plan is as per the project design document.
- The baseline scenario and assessment.
- Enquire/observe the operationality of the systems
- Quality control and quality assurance procedures in place that is applicable for the crediting period.
- Impacts of the project (social, environmental, economic) and ongoing stakeholder feedbacks and approach of addressing those.
- Applicability of the methodology and the parameters that are fixed ex-ante and parameters to be monitored.
- A review of information flows for generating, aggregating and reporting the monitoring parameters

EcoLance Private Limited team has confirmed that the project is implemented in line with the project description and the monitoring system is in line with the VCS-PD.

2.5 Resolution of Findings

The objective of this phase of the validation is to resolve any outstanding issues which need to be clarified for EPL's positive conclusion on the project description. The protocol shows in a transparent manner the requirements, means of verification and the results from verifying the identified criteria. To guarantee transparency any findings raised during the verification are incorporated in the Appendix B to this report. A corrective action request (CAR) is raised if one of the following occurs.

- Non-conformities of the monitoring plan or methodology applied are found in project description that influences the ability of the project activity to achieve real emission reduction if the evidence provided to prove conformity is insufficient;
- Mistakes have been made in applying assumptions, data or estimations of emission reductions that will impair the estimate of emission reductions during crediting period;
- Issues identified in a FAR during previous validation or previous verifications to be validated during crediting period renewal.
- The VCS requirements have not been met and need to correct them.

A clarification request (CL) is raised if information is insufficient or not clear enough to determine whether the applicable VCS requirements have been met.

For this project activity, 02 CLs and 08 CARs were raised during this validation. All CARs and CLs were closed prior to issuance of positive validation opinion. The detail of the findings and their resolution are given under Appendix B of this report.

2.5.1 Forward Action Requests

No Forward Action request has been raised during validation of the project activity.

3 VALIDATION FINDINGS

3.1 Project Details

The project activity utilizes a renewable energy (solar) to generate electricity and supply to grid. It thereby reduces the anthropogenic Green House Gas (GHG) emission that would have been generated in fossil fuel dominated grid. The PD applied emission reduction quantification methodology, ACM0002 version 21.0 /07/ and corresponding associated tools

/12/,/13/. The project is assessed based on the VCS requirements /03/, /04/, /05/, /06/ & UNFCCC criteria of CDM /38/. The assessment on the different features and their compliance with the requirements are given below.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Audit history	Validation by EcoLance Pvt. Ltd. for 1 st crediting period 29-July-2023 to 28-July-2030
Sectoral scope	01 Energy industries (renewable / non-renewables sources)
AFOLU project category, if applicable	This is a non-AFOLU project
Project activity type	Type 1: Renewable Energy Projects
General eligibility of the project to participate in the VCS Program	As per the latest applicable United Nations Datasets /39/, Malawi is a Least Developed Country, therefore a persuasive case is made that the project is eligible under the scope of the VCS Program Standard (V4.7) with reference to Section 2.1.1, 2.1.2 and 2.1.3 namely: • It results in CO₂ emission reductions, one of the seven Kyoto Protocol greenhouse gases. • It consists in grid-connected electricity generation using a greenfield solar power plant in a Least Developed Country • The project activity follows the approved methodology's definition of project area and boundary. The current project is a single installation of greenfield solar PV of 21 MWac capacity. • The project is a greenfield solar PV project in an LDC with very low energy access, therefore the project has been developed with the purpose of creating renewable energy for injection into the Malawian Grid. As Malawi is Least Developed Country as designated by the United Nations, the project does not belong to the projects excluded in Table 1 of VCS Standard.

	Further, the project's eligibility and applicability under the ACM0002 Methodology is explained in detail under Section 3.2 of this Project Design Document.
AFOLU project eligibility, if applicable	This is a non-AFOLU project.
Transfer project eligibility, if applicable	The project has applied only for registration under VERRA VCS. Further, no evidence is found for the rejection or participation of the project activity in any other GHG program such as Gold Standard, UNFCCC CDM /08/,/09/. VVB has conformed through official declaration that they have no intension to register the program in other GHG programs /33/. Same was discussed during the on-site assessment with PP. So, VVB confirm that the project activity is applicable for issuing credit in VERRA VCS and is not rejected by any other GHG programs.
Project design	The project activity is not a grouped project and hence the eligibility criteria for the grouped project is not applicable. The project is categorized as 'Project' with estimated emission reductions of less than or equal to 300,000 tCO ₂ e per year as per section 3.10.1 of VCS standard 4.7 /04/.
Project ownership	Serengeti Nkhotakota Limited owns the project, and is rightly indicated in the PD /01/. This has been cross-checked during the field assessment and from commissioning certificate /16/ & power purchase agreement copy /17/. Therefore, the ownership of the project is with Serengeti Nkhotakota Limited in line with para 3.7.1 of the VCS standard version 4.7 /04/.
Project start date	The project start date is 29th July 2023 which is the date of commissioning of the project activity /16/. From this date the project started generating emission reductions and hence meet the requirement of the definition of start date as stated in VCS standard version 4.7 /04/.
Project crediting period	As stated in the PD /01/, the crediting period start date is 29-July-2023 which is also the project start date. A renewable crediting period with 7 years (twice renewal) has been chosen in line with section 3.9.1 of VCS standard version

	4.7 /04/. So, the first crediting period spans from 29-July-2023 to 28-July-2030 for 7 years.									
Project scale	The total estimated emission reduction during the first 7 years of crediting period is 155,036 tCO₂eq with average annual estimate of the GHG emission reduction for the project activity for the first crediting period as 22,148 tCO₂eq. This has been cross-checked in ER calculation sheet /02/ and the assumption of parameters for the same by PP and found that the assumption of parameters and the ER estimation is without material error. The project scale is categorized as “project” as per section 3.10.1 of VCS standard Version 4.7 /04/. The para 3.10.1 of the VCS standard /04/ categorizes the project activity as “Project” if emission reduction is less than or equal to 300,000 tonnes of CO₂e per year. So, in line with the VCS standard version 4.7 /04/, this project activity is categorized as “Project” and is appropriately indicated in the PD.									
Likelihood of achieving estimated GHG emission reduction or removals	The emission reduction forecast has been checked and it is deemed likely that the stated amount is achieved given that the underlying assumptions do not change.									
Technologies and measures implemented by the project activity or activities	The project activity involves implementation of solar power technology (PV) to generate electricity. A total of 58,458 PV panels each of 445 Wp capacity are installed for 26 Wp installed capacity. This is then converted from DC to AC using 132 string inverters of Huawei SUN2000-105KTL-H1 make. The electricity collected from all the inverters is stepped up to 66 kV by using Step Up Transformer. It is then exported to the national grid through a metering system. The major technical details of the project are given below: <table><tr><th>Technical Parameter</th><th>Value</th><th>Specification</th></tr><tr><td>Solar PV Panels (Sun tech Hi Power STP 445S)</td><td>58,458</td><td>445 Watt per panel (25-year lifespan)</td></tr><tr><td>Inverter (Huawei SUN2000-105KTL-H1)</td><td>132</td><td>105 KW max voltage output</td></tr></table>	Technical Parameter	Value	Specification	Solar PV Panels (Sun tech Hi Power STP 445S)	58,458	445 Watt per panel (25-year lifespan)	Inverter (Huawei SUN2000-105KTL-H1)	132	105 KW max voltage output
Technical Parameter	Value	Specification								
Solar PV Panels (Sun tech Hi Power STP 445S)	58,458	445 Watt per panel (25-year lifespan)								
Inverter (Huawei SUN2000-105KTL-H1)	132	105 KW max voltage output								

	Plant export capacity	21 MW	20.79 MW at 40°C
	Network Voltage (kV)	66 kV grid voltage	11 kV stepped up to 66kV (with 25 MVA transformer)
	Operating frequency	50 Hz	50Hz
	The project technology was cross checked during on-site assessment, cross checking the PV panel's configuration, the single line diagram and other technical specifications. This is further checked and confirmed from commissioning certificates /16/, PPA /17/ and technical specification documents of PV panel and inverters /19/. The project replaces the fossil fuel-based electricity hence by reducing the greenhouse gas emission i.e. CO₂ stipulated by Kyoto Protocol and use the CDM methodology ACM0002 'Grid-connected electricity generation from renewable sources' (Version 21.0) /07/ for quantification of GHG emission which is approved by VCS.		
Implementation schedule of the project activity or activities	The project commissioned for the full capacity of 21 MWp on 29th July 2023 which has been confirmed as per commissioning certificate /16/. The third-party completion certificate also confirms the project is ready for commercial operation as on 06 September 2023 /22/. Therefore, no further implementation is pending and details are consistent with the PD.		
Project location	The project site is located approximately 4km via road to the northwest of Nkhotakota, Malawi. The project site's coordinates are 12°57'50.4"S 34°15'19.9"E. This was validated during on-site audit by the validation team. Hence, project location is correct.		
Conditions prior to project initiation	The project is a green-field project. In the absence of the project activity the equivalent amount of electricity that is generated from the project would have been generated in grid which is primarily fossil fuel dominated. The grid emission factor of the Malawi has been checked to confirm the grid energy mix /40/. Hence, the pre-project scenario is the continuation of current practice.		

Project compliance with applicable laws, statutes and other regulatory frameworks	The project is a voluntary initiative by the project proponent and has not been implemented to meet any local/national laws or regulatory compliances. The project activity is in compliance with current laws and regulations and there are no legal and/or regulatory requirements that prevent the project implementation. The energy sector in Malawi is governed by the Energy Regulation Act /23/, Electricity Act /24/, and the IPP Framework /25/. Renewable Energy is defined in the Energy Regulation Act /23/. The applicable laws do not mandate to set up power plants from solar energy. The validation team has also confirmed that there is no such compliance requirement with an emission trading program or any binding limits on GHG emissions for the project activity in Malawi for this type of particular project. PP has also declared that the emission reduction credits generated by the project activity are/will not be used for any GHG emission targets or binding limits /33/. In addition, VVB checked the commissioning certificate /16/, power purchase agreement copy /17/ and other regulatory approvals /41/ and confirm that the project meets all applicable laws and regulations.
Double counting and participation under other GHG programs	The project has applied only for registration under VERRA VCS. Further, no evidence is found for the rejection of the project activity by any other GHG program when checked in other GHG registries such as UNFCCC CDM /08/, Gold standard /09/, Global Carbon Council /18/ etc. VVB has conformed through official declaration that PP has no intension to register the program in other GHG programs /33/. Same was discussed during the on-site assessment with PP. Therefore, VVB confirm that the project activity is applicable for issuing credit in VERRA VCS and is not rejected by any other GHG programs. Also, PP has no intension to register the project in other GHG program.
No double claiming with emissions trading programs or binding emission limits	EPL validation team confirm that the net GHG emission reduction or removals generated by the project will not be used for compliance with an emission trading program or to meet binding limits on GHG emissions in any Emission Trading Program or other bindings limits.

No double claiming with other forms of environmental credit	The project has no intend to generate any other form of GHG related environmental credit for GHG emission reductions or removals claimed under the VCS program.
Supply chain (Scope 3) emissions double claiming	The project is a grid connected greenfield Solar PV plant, hence scope 3 emissions within the supply chain are not valid under section 3.24.7 of the VCS V4.7, therefore there are no scope 3 emissions associated with this project.
Sustainable development contributions	The project has positive impacts on SDG 7, SDG 8 and SDG 13. As the project has contributed to increase the proportion of clean energy sources, this will contribute to SDG 7. The project will generate job in the sector by contributing to the SDG 8 and the project has resulted emission reduction that contributes to the SDG 13. So, EPL confirm that project will continue to contribute at least to these SDGs that are listed in PD.
Additional information relevant to the project	During the validation of the project activity, VVB confirm that the information provided are not the commercially sensitive information but PP has disclosed all the information transparently. The project uses the solar energy for electricity generation, no leakage management is applicable. Based on the assessment VVB is able to confirm that the project description provided by PP in the PD are accurate and complete in line with the guidance in VCS PD template /05/. PD /01/ has sufficiently provided the information that helps in understanding of the nature of the project activity.

3.2 Safeguards and Stakeholder Engagement

3.2.1 Stakeholder Engagement and Consultation

3.2.1.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	VVB from document review and invitation letters of stakeholder consultation process, noted the below category of people were identified for stakeholder consultation:

	• Department of Environmental Affairs (DEA) • Malawian Investment Trade Centre (MITC) • Nkhotakota District Commissioner's (DC) Office • Representative from Nkhotakota District Lands Office • Local Traditional Authority (TA) • Senior Group Village Head (SGVH) (Head of the Village Heads) The identified stakeholders directly or indirectly impact the project or as they may be impacted by the project activities. VVB confirms the identification of stakeholders are correct for the type of the project and meets the local regulation requirements.
Legal or customary tenure/access rights	Serengeti Nkhotakota Limited owns the project and holds the legal and customary right as verified from commissioning certificate /16/ and power purchase agreement /17/. The plant is set up under government land and VVB checked the land rent receipts to confirm the same /42/. The audit team checked the agreement between Serengeti Nkhotakota Limited & South Pole Carbon Asset Management Ltd. /26/ and noted South Pole Carbon Asset Management Ltd. is acting as consultant for securing and trading of carbon credit to be generated from the project activity. Therefore, the details of project proponent and other entity are transparently detailed in the PD. Hence, are no legal or customary tenure access rights for local stakeholders identified.
Stakeholder diversity and changes over time	Since Serengeti Nkhotakota Limited legally owns and manages the project activity with consultation provided by South Pole Carbon Asset Management Ltd. Therefore, it does not affect stakeholder diversity or changes over time.
Expected changes in well-being	Serengeti Nkhotakota Limited legally owns and manages the project activity, hence, no negative changes in stakeholder well-being are expected.
Location of stakeholders	The stakeholders are from the local community reside in the nearby area of the project location and concerned government entities. The records of those stakeholders were verified during the on-site

	visit, stakeholder attendance sheet and regulatory approvals for the project /16/,/17/.
Location of resources	Since the project activity is carried out by Serengeti Nkhotakota, which also legally owns and manages the project activity, hence, no stakeholder has any access to the project resources. The project users solar energy.

3.2.1.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder engagement process	The project proponent organized initial local stakeholder consultation on 17th January 2018, at Nkhotakota, between representatives of the local District Commissioner (DC), MITC representatives, the local Traditional Authority (TA), Senior Group Village Head (SGVH) and the local Village Heads. This was also attended by the Project Proponent team. The same has been verified during the on-site audit, through record, pictures of public meeting & field operations, and interview held with stakeholders by EcoLance verification team.
Consultation outcome	The project participant conducted the stakeholder consultation in three different modes. They organised a public meeting, Focus group discussions and Field observation. The overall objective was to appraise the identified stakeholders regarding the project activity and take feedback related to economic, social and environmental aspects. The outcome did not result any change in project design but set up a continuous grievance handling procedure which was found effectively in place as detailed in the PD. All relevant documents were made available to the audit team. The same also confirmed from stakeholders during on-site interview. No negative comment or grievance found recorded during the stakeholder meeting. Details of stakeholder consultation is transparently explained in section 2.2 of the PD /01/.
Ongoing communication	As part of ongoing communication, Stakeholders can raise comment via village head which shall be shared with project proponents. In addition, a register is maintained at the gate of plant where

	stakeholder can register their grievance or feedback and the same is reviewed by project proponent. Apart from that the project participants have also conducted a second time LSC meeting and the communities also agreed on livelihood restoration opportunities. This has been checked by the validation team during the on-site inspection interviews. Hence VVB deemed the local stakeholders ongoing communication as appropriate.
Stakeholder input	Validation team is of the opinion that the stakeholder meeting was adequate and appropriate. There were no negative comments raised by the stakeholders and they were totally in support for setting up of the project in the region.

3.2.1.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Obtaining consent	The project is solely owned and operated by Serengeti Nkhotakota Limited. Commissioning certificate /16/ and power purchase agreement /17/ confirms the project has received necessary consents. In addition, the project is set up on a leased land and leased rent receipts /42/ confirms the same.
Outcome of FPIC discussion	The project specifically does not require any FPIC as the ownership lies with project developer i.e. Serengeti Nkhotakota Limited. The project is set up on a leased land for which free, prior and informed consent from land owner has been received as verified from land lease agreement copy and rent payment receipts /27/,/42/.

3.2.1.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Development process	In line with the outcomes of the LSC, the PP has installed grievance boxes at the project site and the nearby communities from where the stakeholders were identified. The same has

	been verified during the on-site audit and noted due records are available with actions in place.
Grievance redress procedure	Grievance boxes have been installed at public spaces where aggrieved stakeholders can anonymously deposit their grievance. These boxes are checked every week and any issue identified is investigated by the management of the project and steps taken to rectify the situation. During on-site audit, the audit team found the grievance recording boxes are put at appropriate places and record keeping and grievance handling register is maintained.

3.2.1.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
No comment received.	Not applicable	VVB noted the project was listed for public comment from 17/08/2023 to 16/09/2023 under VERRA VCS registry. No comment found recorded during the commenting period.

3.2.2 Risks to Local Stakeholders and the Environment

3.2.2.1 Management Experience

From review of publicly available details of the project proponent /45/, VVB confirms the management is experienced in renewable energy projects in the region and has strong commitment towards environment, social and sustainable development. The project has all required regulatory approvals in place.

3.2.2.2 Risk Assessment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human-induced risks to stakeholders' wellbeing	There is no natural & human-induced risk due to the project to stakeholder's wellbeing. The project uses solar energy to generate electricity. The project is set up in a

	unused barren land. The project is benefiting local people in providing jobs and access to clean energy.
Risks to stakeholder participation	No risks identified as this is a solar PV project. VVB while performing the on-site visit found it to be true.
Working conditions	A Health Safety & Environment (HSE) team is present at the project location to monitor & provide satisfactory resolution to any grievance or issue. The same has been verified by the VVB during the audit. PPE kit is worn by operators at workplace. HSE Officer was present and interviewed during on-site audit. Hazardous waste, e-waste are stored in dedicated place with proper label. The project has received Certificate of registration of workplace occupational safety, health and welfare, on 14 February 2023 /41/.
Safety of women and girls	The PP employs women and the project area is well constructed & protected with an established security system. Hence safety of women and girls are given priority by the PP. This has been verified by the VVB. The project has received Certificate of registration of workplace occupational safety, health and welfare, on 14 February 2023 /41/.
Safety of minority and marginalized groups, including children	The project activity does not employ any child labor and there were no minority or marginalized groups associated. The VVB's audit team has verified this during their on-site visit to the project location. Hence, it is not applicable. The project has received Certificate of registration of workplace occupational safety, health and welfare, on 14 February 2023 /41/.
Pollutants (air, noise, discharges to water, generation of waste, release of hazardous materials, and chemical pesticides and fertilizers)	The project being a solar power project based on PV technology, does not emit any pollutions. Intermittently, hazardous oil released during maintenance is stored in dedicated storage system and disposed as per national law. For solar panels, the Life of the solar PV panels is more than 25 years. Although any damaged panel are stored and disposed as per national law.

The project has received Certificate of registration of workplace occupational safety, health and welfare, on 14 February 2023 /41/.

3.2.3 Respect for Human Rights and Equity

3.2.3.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination	The project has received Certificate of registration of workplace occupational safety, health and welfare, on 14 February 2023 /41/. Therefore, no discrimination and sexual harassment policy is in place.
Sexual harassment	The project has received Certificate of registration of workplace occupational safety, health and welfare, on 14 February 2023 /41/. Therefore, no discrimination and sexual harassment policy is in place.
Equal pay for equal work	The project employees both women and men as verified from employment records /46/. Therefore, gender equality in labor and work is in place.
Gender equity in labor and work	The project employees both women and men as verified from employment records /46/. Therefore, gender equality in labor and work is in place.
Forced labor	The project does not have involve forced labor and human traffic.
Child labor	The project does not have involve forced labor and human traffic.
Human trafficking	The project does not have involve forced labor and human traffic.

3.2.3.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No Risk	The company has policies to protect human rights. In addition, Malawian Constitution adopts provisions to protect human rights.

3.2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
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No risk identified.	The project is set up on a leased land and does not harm any protected or cultural heritage. The project has received all applicable regulatory permissions.
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3.2.3.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
No risk identified	Since the project activity is set up and operated by Serengeti Nkhotakota Limited, who legally owns and manages the project activity. The project is set up on a leased land for which free, prior and informed consent from land owner has been received as verified from land lease agreement copy and rent payment receipts /27/,/42/. The project users solar energy.

3.2.3.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Process used to design the benefit sharing plan	Serengeti is sole owner of this project; therefore, no benefit sharing is applicable. This has been checked from the commissioning certificate /16/ and PPA of the project activity /17/.
Summary of the benefit sharing plan	Not applicable since there is no benefit sharing plan.
Approval and dissemination of benefit sharing plan	Not applicable since there is no benefit sharing plan.

3.2.4 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
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Impacts on biodiversity and ecosystems	The project is located in a single location, since this is a solar PV project there is no scope for any kind of direct or indirect impact on the biodiversity and ecosystem of the area. The same has been verified by the VVB during audit.
Soil degradation and soil erosion	Since the project is a solar PV project, hence there is no scope for soil degradation and soil erosion.
Water consumption and stress	No risk, Water Abstraction License has been issued to the PP and same has been verified by the VVB /48/.

3.2.4.1 Rare, Threatened, and Endangered species

Risk identified	Evidence gathering activities, evidence checked, and assessment conclusion
Species and habitat	The project is not located in or adjacent to a habitat of rare, threatened and endangered species. This was confirmed during the site visit and from review of EIA report /44/.
Areas needed for habitat connectivity	Not applicable.

3.2.4.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
N/A	Not applicable since the project is a solar power project.

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
N/A	Not applicable since the project is a solar power project.

	Evidence gathering activities, evidence checked, and assessment conclusion
Invasive species	Not applicable since the project is a solar power project.

3.2.4.3 Ecosystem conversion

Risks Identified	Evidence gathering activities and evidence checked
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Ecosystem conversion:

No risk identified.

The project is a solar power project which does not require any conversion of eco-system. The plant is set up in a barren land as verified during site visit.

3.3 Application of Methodology

3.3.1 Title and Reference

The project correctly applied CDM methodology ACM0002 'Grid-connected electricity generation from renewable sources' version 21.0 /07/.

In addition, below tools are applied in the project activity applicable as per the applied methodology:

TOOL07: Tool to calculate the emission factor for an electricity system, version 07 /13/

TOOL01: Tool for the demonstration and assessment of additionality, version 07 /12/.

Assessment team has checked the UNFCCC website and confirmed that the selection of the applied methodology and methodological tools have been appropriately used and referred.

3.3.2 Applicability

All applicability conditions of applied methodology ACM0002 version 21.0 /07/ are justified in the VCS-PD and VVB's assessment are given below:

Methodology ID	Applicability condition	Assessment and conclusion
ACM0002	This methodology applies to grid-connected renewable energy generation project activities that include: a) Construction and operation of a Greenfield power plant; or b) <i>Retrofitting, rehabilitation (or refurbishment), replacement or capacity addition of an existing power plant.</i>	The project is a greenfield grid-connected solar power-based electricity generation project. VVB has cross-checked the same during field-audit and from review of commissioning certificate /16/. Applicability condition justified.
	<i>Further, the methodology applies to grid-connected renewable energy generation project activities which integrate Battery Energy Storage System (BESS) to a Greenfield power plant or to an existing solar photovoltaic or wind power plant</i>	The project does not integrate Battery Energy Storage System (BESS). VVB has cross-checked the same during field-audit and from review of commissioning certificate /16/. <i>Applicability condition justified.</i>
	In case the project activity involves the integration of a BESS, the methodology is applicable to grid-connected renewable energy power generation project activities that: (a) Integrate BESS with a Greenfield power plant; (b) Integrate a BESS together with implementing a capacity addition to (an) existing solar	The project does not integrate Battery Energy Storage System (BESS). VVB has cross-checked the same during field-audit and from review of commissioning certificate /16/.

	photovoltaic¹ or wind power plant(s)/unit(s); (c) Integrate a BESS to (an) existing solar photovoltaic or wind power plant(s)/unit(s) without implementing any other changes to the existing plant(s); <i>Integrate a BESS together with implementing a retrofit of (an) existing solar photovoltaic or wind power plant(s)/unit(s).</i>	
	The methodology is applicable under the following conditions: (a) Hydro power plant/unit with or without reservoir, wind power plant/unit, geothermal power plant/unit, solar power plant/unit, wave power plant/unit or tidal power plant/unit; (b) In the case of capacity additions, retrofits, rehabilitations or replacements (except for wind, solar, wave or tidal power capacity addition projects) the existing plant/unit started commercial operation prior to the start of a minimum historical reference period of five years, used for the calculation of baseline	The project is a greenfield project; however, does not integrate BESS. VVB has cross-checked the same during field-audit and from review of commissioning certificate /16/.

¹ In case of retrofit or capacity addition for concentrated solar power projects, stakeholders may submit a request for revision to this methodology, providing an apportioning approach to calculate the project emissions due to any fossil fuel consumption attributed to the increased electricity generation from the BESS.

	emissions and defined in the baseline emission section, and no capacity expansion, retrofit, or rehabilitation of the plant/unit has been undertaken between the start of this minimum historical reference period and the implementation of the project activity; (c) In case of Greenfield project activities applicable under paragraph 5 (a) above, the project participants shall demonstrate that the BESS was an integral part of the design of the renewable energy project activity (e.g. by referring to feasibility studies or investment decision documents); The BESS should be charged with electricity generated from the associated renewable energy power plant(s). Only during exigencies² may the BESS be charged with electricity from the grid or a fossil fuel electricity generator. In such cases, the corresponding GHG emissions shall be accounted for as project emissions following the requirements under section Error! Reference source not found.. The charging using the grid or using fossil fuel electricity generator should not amount to more than 2 per	
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² For example, upon deep discharge of the batteries.

	cent of the electricity generated by the project renewable energy plant during a monitoring period. During the time periods (e.g. week(s), months(s)) when the BESS consumes more than 2 per cent of the electricity for charging, the project participant shall not be entitled to issuance of the certified emission reductions for the concerned periods of the monitoring period	
	In the case of integrated hydro power projects, project participants shall: Demonstrate that water flow from upstream power plants/units spill directly to the downstream reservoir and that collectively constitute to the generation capacity of the integrated hydro power project; or Provide an analysis of the water balance covering the water fed to power units, with all possible combinations of reservoirs and without the construction of reservoirs. The purpose of water balance is to demonstrate the requirement of specific combination of reservoirs constructed under CDM project activity for the optimization of power output. This demonstration has to be carried out in the specific scenario of water availability in different seasons to optimize the water flow at the inlet of power units. Therefore, this water balance will take into account seasonal flows from river, tributaries (if any), and rainfall for minimum of five years prior	The project is not a hydro power project. VVB has cross-checked the same during field-audit and from review of commissioning certificate /16/.

	to the implementation of the CDM project activity.	
	The methodology is not applicable to: (a) Project activities that involve switching from fossil fuels to renewable energy sources at the site of the project activity, since in this case the baseline may be the continued use of fossil fuels at the site; (b) Biomass fired power plants/units.	The project activity is a grid connected solar power project and does not involve switching of fuel or biomass. VVB has cross-checked the same during field-audit and from review of commissioning certificate /16/.
	In the case of retrofits, rehabilitations, replacements, or capacity additions, this methodology is only applicable if the most plausible baseline scenario, as a result of the identification of baseline scenario, is “the continuation of the current situation, that is to use the power generation equipment that was already in use prior to the implementation of the project activity and undertaking business as usual maintenance”.	The project is a greenfield project. VVB has cross-checked the same during field-audit and from review of commissioning certificate /16/.
	In addition, the applicability conditions included in the tools referred to below apply.	<i>The project correctly applies the below tools:</i> • <i>TOOL07: Tool to calculate the emission factor for an electricity system, version 07</i> TOOL01: Tool for the demonstration and

		assessment of additionality, version 07
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In view of the above consideration, the assessment team is able to conclude that the used methodologies and tools are applicable for the project activity.

3.3.3 Project Boundary

As per the methodology 'the spatial extent of the project boundary includes the project power plant/unit and all power plants/units connected physically to the electricity system that the CDM project power plant is connected to'. The generated electricity will be delivered to the Malawian National Grid through the connected sub-station. Emissions sources included in the project boundary are shown in the table below.

Emission sources which are not addressed by the applied methodology and which are expected to contribute more than 1% of the overall expected average annual emissions reduction have not been identified because the project activity is a Greenfield solar power project and involves mainly assembly and erection of pre-fabricated components.

From the assessment of VVB, PP has appropriately indicated the project boundary and the baseline emission sources, project emission sources and leakage emissions and the information stipulated in the updated PD /01/ are justifiable and valid.

Source		Gas	Included	Justification/explanation
Baseline	CO ₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity	CO ₂	Yes	Main emission source
		CH ₄	No	Minor emission source
		N ₂ O	No	Minor emission source
Project activity	The project does not have any project emissions			

3.3.4 Baseline Scenario

As per the applied methodology ACM0002, version 21.0 the baseline is 'If the project activity is the installation of a Greenfield power plant with or without a BESS as described under paragraph 4(a) or paragraph 5(a), the baseline scenario is electricity delivered to the grid by

the project activity that would have otherwise been generated by the operation of grid-connected power plants and by the addition of new generation sources, as reflected in the combined margin (CM) calculations described in TOOL07'.

Since, the project is a greenfield project without BESS, the baseline is adequately described in the PD. The combined margin (CM) calculations described in TOOL07 has been referred and credible and authentic source are referred as discussed in section 4.1 of the PD and section 3.3.6 of this report. From the VVB's assessment, it can be concluded that the identified baseline scenario is justified and according to the VCS Standard version 4.5 /04/.

3.3.5 Additionality

The project proponent has demonstrated additionality of the project as per para 5.3.2 applying TOOL 1 'Tool for the demonstration and assessment of additionality' version 7 /12/.

A step-wise approach has been followed as per the tool. **Step 0** is not applicable as the project is not a first-of its kind project.

Step 1: Identification of alternatives to the project activity:

while identifying baseline alternatives, two alternatives identified as follows:

- a) The proposed project activity undertaken without being registered as VCS project activity
- b) Continuation of existing situation

The energy sector in Malawi is governed by the Energy Regulation Act /23/, Electricity Act /24/, and the IPP Framework /25/. Renewable Energy is defined in the Energy Regulation Act /23/. The above mentioned regulations does not influence the choice of fuel used for power generation. There is no legal requirement on the choice of a particular technology for power generation. Therefore, alternatives identified are in consistent with mandatory laws and regulations.

Step 2: Investment analysis has been applied to demonstrate additionality in line with the additionality tool /15/.

Choice of approach:

The PP has chosen to apply the benchmark analysis method and has identified post tax equity IRR as the most suitable financial indicator. The project cannot apply simple cost analysis since the project brings revenue from the sale of electricity; also, investment comparison analysis cannot be applied as the alternative to the project activity is the electricity generated by new and existing grid connected power plants. Since, the alternative to the project activity is the supply of electricity from grid which is not to be considered an investment and therefore benchmark approach is considered appropriate. Since the project proponent is demonstrating the financial unattractiveness of the project, the equity IRR is

appropriate, as the project cost involves both equity and debt. Hence, equity IRR is considered by the PP and is appropriate for the project activity.

Benchmark selection:

In accordance to “investment analysis tool” /15/, para 17, “in the case of projects which could be developed by an entity other than the project participant, then the benchmark should be based on the parameters that are standard in the market”. The tool recommends country and project specific expected returns on equity in Appendix of the tool. The tool also states that in situations where an investment analysis is carried out in nominal terms, and available IRR benchmarks are in real terms, project participants shall convert the real term values of benchmarks to nominal terms by adding the inflation forecast of the central bank of the host country for the duration of the crediting period. If this information is not available, then the average forecasted inflation rate for the host country published by IMF or the World Bank for the next five years after the start of the project activity shall be used

PP has finalized power tariff at the time prior to project commission and considered inflation for O & M cost and other variables; hence, the equity IRR is calculated in nominal terms. Accordingly, PP considered default expected return on equity for energy industries (Group 1) applicable for Malawi as 24.01% applicable for group 1 projects (as per version 13 of the tool which is the latest version of the tool at the time of validation). The inflation rate has been taken as average of forecast for years (2023-2026) by IMF as 5.65% has been taken /43/. Accordingly, the resulted benchmark after converting to nominal terms is 31.02%.

Input Parameters:

The validation team validated the input values and assumptions in the investment analysis by checking the original and other supportive documents as detailed below. It is noted that input values stated in the VCS-DD are consistent with that of the financial calculation sheet /29/. The lifetime of the project is confirmed to be 25 years as per the technology supplier /19/, and the investment analysis is done for the period of 25 years and hence is justified as per the investment analysis tool /15/:

Following major timelines are consider as part of assessment of investment analysis:

Milestones	Date
Company incorporation	2 May 2018 /21/
Environmental Impact Assessment	29 June 2019 /44/
Loan Ccommitment Letter	20 December 2018 /20/
Third party due diligence report (DPR)	January 2019 /30/
Signing of Power Purchase Agreement	14 February 2019 /17/

EPC Contract	14 April 2020 /51/
O & M Contract	9th June 2020 /49/
Commissioning of the project	29 July 2023/16/

EcoLance has validated the input parameters used in the investment analysis and the following steps have been followed to assess the investment analysis:

- Assessment of the sources used for input parameters. All input parameters used in the financial analysis are taken from Due Diligence Report /30/, PPA /17/, Loan Sanction Letter /20/ etc. as described in the following table.
- Confirmation of the values in the PD and investment analysis is fully consistent with the values provided in source documents. VVB compared the input parameters for the financial analysis included in the VCS-PD /02/ and in the investment analysis spreadsheet /29/ with the parameters stated in the documents used and was able to confirm that the values applied are consistent with the values stated source documents.

Cross-check of the input parameters used in the financial analysis. The input parameters used in the financial analysis were cross-checked with sources used. The following is carried out.

Parameter	Value	Source	Validation Assessment and cross checking
Project Capacity	21 MW	As per PPA, DPR, Loan sanction letter	Verified against the loan sanction letter /20/, The PPA /17/ and commissioning certificate /16/. Further, on-site audit confirms the project capacity.
Expected electricity generation	56,458 MWh	Third Party assessment Report /30/	Verified against the due diligence report prepared by Mott MacDonald /30/ prior to execution of PPA. Therefore, it represents credible in decision making to go ahead with the project.
Annual degradation	0.45%	Third Party assessment Report /30/	Verified against the due diligence report prepared by Mott MacDonald /30/ prior to execution of PPA. Therefore, it represents credible in decision making to go ahead with the project.
Project cost	45.80 USD million	As per audited balance sheet /47/	The actual project cost as per audited balance sheet of the project is taken for project /47/. Whereas, the project cost as per loan approval letter is 66.60 USD million /20/. The loan

			approval was prior to signing of PPA which was the basis of decision making. However, on conservative side, PP has taken the actual value for showing the IRR value. This is conservative and hence, the project cost considered is found to be correct.
Debt	75% of project cost	As per loan approval letter /20/	The loan approval from Overseas Private Investment Corporation /20/ prior to execution of PPA. As per the loan commitment letter, the debt is 75% of project cost.
Equity	25% of project cost	As per loan approval letter /20/	
Interest Rate	3.5%	As per loan approval letter /20/	As per loan approval letter the interest rate is 3.5% which is acceptable /20/.
Tenure of term loan	17 years	As per loan approval letter /20/	The tenure of term loan is considered by PP at the time of investment decision as per loan commitment letter /20/.
Tariff	USD 0.0972/kWh (fixed for 20 years)	Tariff rate is as per PPA executed /17/	The tariff rate is as per PPA /17/. Hence, actual tariff rate is considered for IRR analysis. VVB checked actual invoices raised post commissioning and noted the rate to be same. Hence, accepted.
Corporate tax rate	15%	As per tax law in Malawi /50/	The tax rate is cross checked and found to be correct as per host country law /50/.

VVB thus confirm that all input data, assumptions used in the investment analysis are credible for analysis of equity IRR. The validation team cross checked all data parameters against credible sources (where available public sources and third-party reports) as discussed above and confirm that the values used in the VCS-PD and investment analysis are consistent with the sources used for the input parameters. The accounting principles adopted with respect to computation are found to be in order. The arithmetical accuracy is also found to be correct.

The equity IRR calculations /29/ was provided in excel spreadsheet and verified. The assumptions used in the calculations were deemed to be correct and verified by VVB. The equity IRR of the project activity without carbon revenue is 12.78% which confirms that the proposed project activity in absence of carbon benefits as compared to benchmark of 31.02% is not financially attractive.

Sensitivity analysis:

The investment analysis tool /15/ requires that the investment analysis should contain a sensitivity analysis that supports the robustness of the conclusion arrived at by varying the

critical assumptions to a reasonable variation ($\pm 10\%$). The project developer has identified the following parameters as the most critical assumptions.

- PLF/Net generation
- Operating Expense
- Project Cost
- Tariff

Results of sensitivity analysis are presented below:

	IRR		
Variation %	-10%	0%	10%
PLF	9.87%	12.78%	16.88%
O & M cost	13.46%	12.78%	12.10%
Project Cost	14.09%	12.78%	11.70%
Tariff	8.87%	12.78%	16.88%

The above analysis shows that the IRR does not cross the benchmark at $\pm 10\%$ variations of the above parameters. The following table reveals the changes/variations required to the parameters for the IRR to cross the benchmark:

Parameter	Changes/Variations	Possibility of the situation
PLF	PLF further increases by 43.10% to reach 35.5% PLF.	The PLF is assessed by third party and value is taken accordingly. As per International Renewable Energy Agency report on Malawi, annual generation per unit of installed PV capacity is around 1.6-1.8 MWh/kWp resulting a PLF of around 21-22% /52/. Therefore, it is highly unlikely to increase the PLF to 35.5%.
O & M Cost	O & M cost reduces by 271%.	O & M contract has been executed as per the cost given in the IRR worksheet which is not expected to reduce by 271%. Due to inflation, it is likely the cost will increase over the years. Hence, this is not a possible scenario.

Project Cost	Project cost reduced by 59.10%	The IRR sheet takes into account the actual cost incurred for the project from the audited balance sheet. Therefore, there is no possibility to reduce the project cost by 59%.
Tariff	Tariff is reduced by 99.50%	The power purchase agreement for the project is already executed for a tariff rate of USD 0.0972/kWh (fixed for 20 years). Therefore, it is unlikely to reduce the tariff by 99.50%.

Step 4: Common Practice Analysis

The PP has conducted the common practice analysis as per the requirements of “Tool for the demonstration and assessment of additionality Version 07.0.0 /12/. In line with the tool, the ‘methodological tool common practice’ version 03.1 has been applied /14/. PP has selected Malawi (host country) as the default applicable geographical area for the assessment of common practice.

As per the Step 1 of the guideline /14/, the applicable output range as $\pm 50\%$ of the design output range (21 MW) is 10.5 MW to 31.5 MW.

In the step 2, PP has identified the plants that deliver the output in the range of 10.5 MW to 31.5 MW in Malawi and identified projects with same measure (power generation based on renewable energy generation) and projects that use the same energy source/fuel and feedstock. Only Solar PV projects considered since the technology measure and tariff structure and policy framework are different for different types of RE technologies which started commercial operation before start date of the project i.e. 14 April 2020. VVB checked publicly available data for Malawi country from Electricity Supply Corporation of Malawi (ESCOM) Limited & Electricity Generation Company (Malawi) Limited (EGENCO) and confirm the details to be correct.

One project ‘Golomoti 20MW Solar PV Project’ found with the capacity range however the same is found listed under VCS with ID 4233 which started operation second quarter of the year 2022 /53/. Hence, no project identified.

In step 3: within the projects identified in Step 2, identify those that are neither registered CDM project activities, project activities submitted for registration, nor project activities undergoing validation. Note their number N_{all} . Since, no project identified in step 2, N_{all} is 0.

In Step 4, within similar projects identified in Step 3, identify those that apply technologies that are different to the technology applied in the proposed project activity. Note their number N_{diff} . From step 3 it is concluded N_{diff} is 0.

Step 5: calculate factor $F=1-N_{diff}/N_{all}$ which results to be 1.

The proposed project activity is a “common practice” within a sector in the applicable geographical area if the factor F is greater than 0.2 and $N_{all}-N_{diff}$ is greater than 3.

F is greater than 0.2 but $N_{all}-N_{diff}$ is less than 3; hence, the project activity is not a common practice in the applicable geographical area.

Hence, the project is additional.

3.3.6 Quantification of GHG Emission Reductions and Carbon Dioxide Removals

As per ACM0002 Version 21.0, equation 11, baseline emissions include only CO₂ emissions from electricity generation in fossil fuel fired power plants that are displaced due to the project activity and are calculated as follows:

$$BE_y = EG_{PJ,y} \times EF_{grid,CM,y}$$

Where:

BE_y	=	Baseline emissions in year y (t CO ₂ /yr)
$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EF_{grid,CM,y}$	=	Combined margin CO ₂ emission factor for grid connected power generation in year y calculated using TOOL07 (t CO ₂ /MWh)

Calculation of $EG_{PJ,y}$:

The project activity being the Greenfield plant, $EG_{PJ,y}$ is as per equation 12 of the methodology is as below:

$$EG_{PJ,y} = EG_{facility,y}$$

Where:

$EG_{PJ,y}$	=	Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh/yr)
$EG_{facility,y}$	=	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh/yr)

Determination of $EG_{\text{facility},y}$: For ex-ante estimation, the amount of electricity delivered to grid is taken as per third party due diligence report as 56,458 MWh/year with 0.45% degradation from second year /30/. For ex-post, the value shall be monitored in calibrated energy meter and as recorded in monthly generation report issued by government entity. The same can be cross checked from the invoice copy raised.

Determination of $EF_{\text{grid,CM},y}$: CM (combined margin) grid emission factor has been calculated on the basis of sum of 75% of OM (operating margin) and 25% of BM (build margin) following tool to calculate emission factor for electricity system /13/.

$$EF_{\text{grid,CM},y} = 0.75 * EF_{\text{grid,OM},y} + 0.25 * EF_{\text{grid,BM},y}$$

The CM emission factor is calculated as 0.3970 tCO_{2e}/MWh sourced from UNFCCC (Harmonized IFI Default Grid Factors 2021 v3.2) /40/. The emission factor is published by International Financial Institutions technical working group on greenhouse gas accounting following 'methodological approach for the common default grid emission factor dataset' /40/. The above methodology clearly states the approach of calculating CM following procedure stipulated in ACM0002 methodology considering Operating Margin (OM) and Build Margin (BM) and following AHSA-001 'IFI Approach to GHG accounting for renewable energy projects, version 2.0' /54/. Therefore, the CM emission factor calculated is as per the 'tool to calculate emission factor for electricity system' /13/ and correctly considered the value to be 0.3970 tCO_{2e}/MWh for Intermittent Energy (e.g., Solar, Wind, Tidal) /40/.

The CM grid emission factor is fixed ex-ante for the entire first crediting period.

Baseline emission estimation:

Ex-ante the baseline emissions are estimated as below:

$$BE_y = EG_{PJ,y} \times EF_{\text{grid,CM},y}$$

Year	Generation (MWh)	Degradation rate	Grid EF	BE _y (tCO _{2e})
1	56,548	0	0.397	22,450
2	56,293	0.45%	0.397	22,348
3	56,039	0.45%	0.397	22,247
4	55,786	0.45%	0.397	22,147
5	55,534	0.45%	0.397	22,047
6	55,284	0.45%	0.397	21,948

7	55,035	0.45%	0.397	21,849
Total	390,519	-	-	155,036
Average	55,788	-	-	22,148

Project Emission estimation:

In line with the applied methodology /07/, no project emission has been considered for the project.

$$PE_y = 0 \text{ tCO}_2\text{eq}$$

Leakage emission:

As per the applied methodology /07/, no other leakage emissions are considered.

$$\text{So, } LE_y = 0$$

Emission Reduction:

In line with para 62 (equation 17) of the applied methodology /07/, the net emission reduction shall be calculated as below:

$$ER_y = BE_y - PE_y$$

Therefore, ex-ante estimated emission reduction is 22,148 tCO₂e/year (average) over the first 7 years of crediting period.

In assessment of VVB, the PD appropriately apply the applied methodology /06/ and the corresponding tools /06/ to estimate baseline emission, project emission and leakage emission to calculate emission reduction by the project activity during the crediting period. The equations used are well complied with the methodology and tools and the assumption for the ex-ante fixed values/parameters are applied correctly without any material error.

3.3.7 Methodology Deviations

VVB confirm that methodology deviation is not applicable.

3.3.8 Monitoring Plan

The monitoring plan is in accordance with the monitoring methodology; the monitoring plan will give opportunity for real measurement of achieved emission reductions. Below are the parameters to be monitored.

	Parameter	Description/Assessment
1	Quantity of net electricity generation supplied by the project plant/unit to the grid in year y $EG_{Facility,y}/EG_{PJ,y}$ (MWh/year)	The net electricity supplied to grid is monitored continuously in a main and back up meter. The meters are two-way and records both energy import and export. The net export is calculated as difference of gross export and import. Monthly recording is maintained. Invoice is raised as per the monthly net generation records. The energy meters are calibrated annually. The monitoring and recording procedure meets the methodology requirements /07/.

The monitoring plan in the PD provides detailed information related to the collection and archiving of all relevant data needed, quality control measures as required by the applied methodologies. The monitoring arrangement and responsible personal as per the monitoring plan found effective during site visit. The monitoring arrangements are feasible within the project design. The PP has the ability to implement the monitoring plan as per the PD /01/.

3.4 Non-Permanence Risk Analysis

The non-permanence risk is not identified and hence is not applicable.

4 VALIDATION OPINION

4.1 Validation Summary

The project participants of the project are responsible for:

- the preparation of greenhouses gas emissions data and the reported greenhouse gas emission reductions estimates for the ex-ante for the crediting period on the basis provided by applied methodology
- updating the baseline in line with the VCS standard 4.7 /04/ and prepare the updated project description in line with the guidance provided in the project description template.
- Provide justification and evidences for the assumption made in the PD /01/.

It is the responsibility of EPL to express an independent validation opinion about the project's conformity with the requirements of VCS Standard version 4.7 and applied methodology on the estimated greenhouse gas emission reductions from the project for the crediting period. Based on documented evidence and corroborated by an on-site assessment, EPL can confirm that:

- the project activity has been implemented and operated as per the Project Description /01/;
- the project description /01/, ex-ante estimates emission reduction sheet /02/ and other supporting documents provided are complete and verifiable and in accordance with the applicable VCS Standard version 4.7 requirements;
- The baseline is correctly identified in line with the VCS Standard version 4.7 requirements and the associated applied methodology and tools. The project correctly applies the approved baseline and monitoring methodology “ACM0002: Grid connected electricity generation from renewable sources (Version 21.0)” /07/.
- Data/parameters used for the ex-ante emission are based on credible source and hence are correctly presented.

The validation of the GHG statements was done in accordance of the 14064-03; 2019. It is EPL’s opinion that the GHG emission reduction stated in the PD (version 04 of 06 February 2025) for the “Nkhotakota Solar PV project” for the crediting period (29-July-2023 to 28-July-2030) and corresponding emission reduction estimation sheet (version 01 of 17-January-2024) are fairly stated. The GHG emission reductions are calculated on the basis of approved methodology ACM0002: Grid connected electricity generation from renewable sources (Version 21.0) and the monitoring plan included in the Project Description is complete and all other relevant UNFCCC and VCS Standard Version 4.7 requirements. The project activity also complies with all relevant host Party criteria and correctly applies the baseline and monitoring methodology.

Hence EPL is able to validate the project and recommend for registration of the project activity for the period validated as shown below with corresponding emission reduction estimates for the crediting period:

Validation period: From 29-July-2023 to 28-July-2030

Validated GHG emission reductions and removals in the above period.

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
Year 1 (29-July-2023 to 28-July-2024)	22,450
Year 2 (29-July-2024 to 28-July-2025)	22,348
Year 3 (29-July-2025 to 28-July-2026)	22,247
Year 4 (29-July-2026 to 28-July-2027)	22,147

Year 5 (29-July-2027 to 28-July-2028)	22,047
Year 6 (29-July-2028 to 28-July-2029)	21,948
Year 7 (29-July-2029 to 28-July-2030)	21,849
Total estimated ERs	155,036
Total number of crediting years	7
Average annual ERs	22,148

4.2 Validation Conclusion

The total emission reductions from the project “Nkhotakota Solar PV project”, Malawi” are estimated to be on average 22,148 tCO₂e per year over the 7 years crediting period and twice renewable. The emission reduction forecast has been checked and it is deemed likely that the stated amount is achieved given that the underlying assumptions do not change.

In conclusion, it is EPL’s opinion that the project activity “Nkhotakota Solar PV project” in Malawi, as described in the VCS PD version 4.0 of 06/02/2025, meets all relevant UNFCCC and VCS Version 4.7 requirements for the VCS VER and all relevant host Party criteria and correctly applies the UNFCCC methodology “ACM0002, version 21.0. EPL thus requests registration of the project as a VCS project activity.

Crediting period: From 29-July-2023 to 28-July-2030

Validated estimated GHG emission reductions and carbon dioxide removals for the project crediting period:

Vintage period	Estimated baseline emissions (tCO ₂ e)	Estimated project emissions (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated reduction VCU (tCO ₂ e)	Estimated removal VCU (tCO ₂ e)	Estimated total VCUs (tCO ₂ e)
29-July-2023 to 31-December-2023	9,228	0	0	9,228		9,228
01-January-2024 to 31-December-2024	22,148	0	0	22,148		22,148
01-January-2025 to	22,148	0	0	22,148		22,148

31-December-2025						
01-January-2026 to 31-December-2026	22,148	0	0	22,148		22,148
01-January-2027 to 31-December-2027	22,148	0	0	22,148		22,148
01-January-2028 to 31-December-2028	22,148	0	0	22,148		22,148
01-January-2029 to 31-December-2029	22,148	0	0	22,148		22,148
01-January-2030 to 29-July-2030	12,919	0	0	12,919		12,919
Total	155,036	0	0	155,036		155,036

APPENDIX 1: COMMERCIALY SENSITIVE INFORMATION

Not applicable.

APPENDIX A: ABBREVIATIONS

BE	Baseline Emissions
CAR	Corrective Action Request
CDM	Clean Development Mechanism
CDM M&P	Modalities and Procedures CDM
CL	Clarification Request
CO ₂	Carbon dioxide
CO ₂ e	Carbon dioxide equivalent
DNA	Designated National Authority
EF	Emission Factor
EIA	Environmental Impact Assessment
EPL	EcoLance Private Limited
ER	Emission Reductions
FAR	Forward Action Request
GHG(s)	Greenhouse gas(es)
GWP	Global Warming Potential
IPCC	Intergovernmental Panel on Climate Change
MoV	Means of Verification
MR	Monitoring Report
PDD	Project Design Document
PP(s)	Project Participant(s)
Ref.	Document Reference
SS(s)	Sectoral Scope(s)
TA(s)	Technical Area(s)
UNFCCC	United Nations Framework Convention on Climate Change
VCU	Verified Carbon Unit
VVS	Validation and Verification Standard
VVB	Validation and verification body

APPENDIX B: CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS AND FORWARD ACTION REQUESTS

Table 1. CL from this verification

CL ID	01	Section no.	General	Date: 24/11/2023
Description of CL				
PP is requested to provide following documents/evidence: 1. Commissioning certificate 2. Project ownership documents/evidence 3. Licence/Permit documents 1) i) Environmental impact Assessment 2) ii) Water Abstraction 3) iii) General waste handling permit 4) iv) Occupational Health and Safety licence 4. Technical Specification 5. Single line diagram and meter locations 6. Layout diagram of role and responsibility of the employee/owner in the project. 7. Official declaration that whether the project activity is included in other emission trading programs or under any emission reduction binding limits and net GHG emission reductions or removals generated during this monitoring period have not been used for compliance under such programs or mechanisms.				
Project participant response				Date: 17/01/2024
The Project Proponent submits the above mentioned documents to fulfil this requirement.				
Documentation provided by project participant				
1. Commissioning certificate 2. Project ownership documents/evidence 3. Licence/Permit documents 5) i) Environmental impact Assessment 6) ii) Water Abstraction 7) iii) General waste handling permit 8) iv) Occupational Health and Safety licence 4. Technical Specification 5. Single line diagram and meter locations 6. Organogram 7. Official declaration from PP that no other GHG program is being participated in				
VVB's assessment				Date: 20/07/2024
The audit team reviewed the submitted documents and ascertain the project design is as per the description provided in the PD. The project is commissioned and all applicable regulatory approvals are in place. Hence, CL is closed.				

CL ID	02	Section no.	Additionality of the project	Date: 24/11/2023
Description of CL				
PP is requested to provide following evidences with respect to IRR analysis: 1. Investment decision date 2. The Detailed Project Report (DPR) 3. All purchase orders/equipment supply contract for major equipments (PV module, Main transformer, Inverters, etc.) 4. Other expenses if any 5. Operation and Maintenance Agreement. 6. Power Purchase Agreements 7. Debt equity ratio and actual loan sanction document				
Project participant response				Date: 17/01/2024
In line with the request above, the PP submits the required documents				
Documentation provided by project participant				
1. Investment decision date 2. The Detailed Project Report (DPR) 3. All purchase orders/equipment supply contract for major equipments (PV module, Main transformer, Inverters, etc.) 4. Other expenses if any 5. Operation and Maintenance Agreement. 6. Power Purchase Agreements 7. Debt equity ratio and actual loan sanction document				
VVB's assessment				Date: 20/07/2024
The chronology of events are transparently presented in the updated PD and supporting documents are provided. The audit team reviewed the documents and confirm the chronology and respective dates are correct. In addition, the post-tax equity IRR has been demonstrated which meets the requirements of UNFCCC additionality tool and computation is done with input data which are valid and correct at the time of investment decision and project commissioning. The detail assessment is presented in the validation report appropriate section, and additionality is justified. Hence, CL is closed.				

Table 1. CAR from this verification

CAR ID	01	Section no.	VCS-PD template requirements	Date: 24/11/2023
Description of CAR				

1. The VCS-PD template is altered in various sections. PP is requested to follow the instructions to complete the PD without altering the project specific requirements. 2. The latest VCS Project Description report template, v 4.3 not used. As per the template the VCS logo and version of the template document is not reflecting. 3. The exact location of the project is not stated in section 1.1 of the PD as per PD completion guideline requirements. 4. Section 1.1 of the VCS-PD template consist of a table showing the audit type, period, duration, etc. which is not available in the submitted PD.	
Project participant response	Date: 17/01/2024
1) The PDD was submitted on 26 th May 2023 whereas the latest template 4.3 was released on 23 rd August 2023. Hence Version 4.2 of the PDD template has been used which is applicable till March 2024. 2) Please refer to response 1 to satisfy this CAR 3) The referred section 1.1 has been updated within section 1.1 of the PD V1.2 4) Please refer to response 1	
Documentation provided by project participant	
1) Updated VCS PD Version 1.2 dated 11 th January 2023	
VVB's assessment	Date: 20/07/2024
The updated PD is completed using latest template v4.4 and completed as per instructions therein. Hence, corrections are found in order and CAR is closed.	

CAR ID	02	Section no.		Date: 24/11/2023
Description of CAR				
1. The project start date is not correct in the VCS-PD. Also describe how the date meets the definition of project start date as per VCS standard. 2. The start and end date of project crediting period needs to be specified with total length as per PD-completion guideline. 3. In section 1.10, the estimated GHG emission reduction table needs to specify the year (e.g. 2019) as per PD completion guideline.				
Project participant response				Date: 17/01/2024
1) In line with the guidance under VCS Standard 4.4, the start date of project commissioning is updated as per latest Grid Connection Certificate issued by ESCOM and attached. 2) The referred section has been updated to correctly reflect the project crediting period 3) The referred section has been updated reflecting the same				
Documentation provided by project participant				
1) ESCOM Grid Connection Certificate 2) Updated VCS PD Version 1.2 dated 11 th January 2023				

VVB's assessment	Date: 20/07/2024
The start date of the project is appropriately justified and considered. The same is as per VCS requirement. Accordingly, the start and end date of crediting period is provided correctly. Other necessary corrections are found in order and hence CAR is closed.	

CAR ID	03	Section no.		Date: 24/11/2023
Description of CAR				
1. The technical component details such as number of PV modules, inverters, model type, specification, generator details etc. are not correct in section 1.11 of the PD as per actual implemented data observed at site. 2. Section 1.13 is not discussed/described as per requirement of PD completion guideline. 3. Section 1.16.3 of the VCS-PD, the discussion on supply chain (scope 3) emissions are not accurate as per the VCS-PD completion guideline. 4. Under No Net Harm, it is not discussed whether the project has any negative socio-economic impacts as the project uses huge land area. Also for environmental aspect, explain whether there will be negative impact from water extraction and consumption for cleaning PV modules.				
Project participant response				Date: 17/01/2024
1) The PD is updated to reflect the technical component details as demonstrated at the On-Site Visit. 2) The Section 1.13 is further elaborated to demonstrate the prior conditions on site in line with the PD Completion Guidelines 3) The referred section is updated to reflect the current status, as the PD template version 4.3 is used. 4) The project proponent has issued a water abstraction license from the Malawian government and the same is submitted. Water consumption for cleaning will be minimal.				
Documentation provided by project participant				
1) Updated VCS PD Version 1.2 dated 11th January 2023 2) Water abstraction license				
VVB's assessment				Date: 20/07/2024
The updated PD included the system components and all monitoring points correctly as per actual arrangement in place. The supply chain (scope 3) emissions are appropriately discussed in the PD. Also, socio-environmental impacts are discussed and VVB could confirm the details are correct as per EIA study and other approvals in place. Updated PD is accepted and CAR is closed.				

CAR ID	04	Section no.		Date: 24/11/2023
Description of CAR				
As per PD completion guideline the followings are not included in section 2.2 of the VCS-PD <i>The procedures or methods used for engaging local stakeholders (e.g., dates of announcements or meetings, periods during which input was sought).</i> <i>The procedures or methods used for documenting the outcomes of the local stakeholder consultation.</i> <i>The mechanism for on-going communication with local stakeholders.</i> - How due account of all and any input received during the consultation has been taken. Include details on any updates to the project design or justify why updates are not appropriate				
Project participant response				Date: 17/01/2024
1) In line with the guidance for LSC under VCS Standard 4.4, the original LSC was carried out on October 2019. Further, in line with the PP's continuous communication and engagement, a second LSC was carried out on 12 th October 2023 in line with the requirements. The dates, invitation and attendance proofs are attached as annexure 2 within the PD. 2) The methods for documenting the outcomes are inline with the guidance under VCS Standard 4.4 and are attached as annexure-2 within the PD. 3) The PP has installed Grievance Registers at public spaces and the project site to ensure the basic needs of the local community members are maintained. These are attached as Annexure 1 within the PD. 4) Please refer to response 2 for the conclusion of this CAR. Therefore it is demonstrated that the PP has ensured incorporation of all LSC inputs				
Documentation provided by project participant				
1) <i>Grievance Register images</i> 2) <i>LSC invitations attendance, MOM and images</i> 3) Updated VCS PD Version 1.2 dated 11th January 2023				
VVB's assessment				Date: 20/07/2024
The stakeholder consultation details, the feedback and comments are appropriately discussed in the PD. VVB could verify the same from stakeholder consultation records and site visit interview. The on-going communication mechanism is detailed as per actual arrangement in place which audit team could confirm during site visit. Hence, the response and corrections are accepted and CAR is closed.				

CAR ID	05	Section no.		Date: 24/11/2023
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Description of CAR	
1. As per PD completion guideline, section 2.3 needs to summarize any environmental impact assessments carried out with respect to the project which is not transparent in the PD. 2. Section 2.4 of the PD does not provide the public commenting period and if any comment received and due actions taken.	
Project participant response	Date: 17/01/2024
1) In line with the guidance provided under VCS Standard 4.4, It is submitted that the project proponent has car 2) The referred section in the PDD has been further updated to reflect the same in more detail. No Comments were received during the VERRA Public Comment Period dated 17th August 2023 to 16th September 2023.	
Documentation provided by project participant	
1) Updated VCS PD Version 1.2 dated 11 th January 2023	
VVB's assessment	Date: 20/07/2024
The updated PD describes socio-environmental impacts and VVB could confirm the details to be correct as per EIA study and other approvals in place. Updated PD is accepted and CAR is closed.	

CAR ID	06	Section no.		Date: 24/11/2023
Description of CAR				
Section 3.2 of the PD does not describe the applicability conditions of applied tools and how the conditions are met by the project activity.				
Project participant response				Date: 17/01/2024
In line with the guidance mentioned in the VCS Standard 4.4 , the PDD is updated to reflect specific tools used.				
Documentation provided by project participant				
1) Updated VCS PD Version 1.2 dated 11 th January 2023				
VVB's assessment				Date: 20/07/2024
The methodology applicability conditions and other applicable tools are appropriately discussed in relevant sections of the PD. VVB provided justification of applicability of applied tools in appropriate section. Response is accepted and CAR is closed.				

CAR ID	07	Section no.		Date: 24/11/2023
Description of CAR				

1. Step 1(b) of additionality analysis in section 3.5 of the PD did not list out the applicable laws/regulations and how identified alternatives are consistent with these laws. 2. While discussing sub-step 2b (apply benchmark analysis), PP has referred version 8 of the Investment Analysis tool which is not the latest version of the tool. PP shall consider the latest version of the tool applicable and valid at the time of validation. 3. The major input values with data source are not listed which are used for the IRR analysis in section 3.5 of the PD. 4. The sensitivity analysis should also discuss the probability at which it reaches the benchmark. 5. Common Practice Step 2 (b) projects that apply the same measure with the project activity, -i.e. power generation based on renewables- shall be considered according to Tool 24. Para (c) is only applicable If a technology switch measure Is Implemented. Because of considering only the solar power plants In Step 2 results under Step 4 automatically that no different technologies can be identified and hence Ndiff Is calculated as 0.	
Project participant response	Date: 17/01/2024
1) It is submitted by the PP that the current laws and policies of Malawi are up to date 2) The version 8 is appropriate as the Client Board's investment decision date is November 2018. For the year of 2018, the Version 8 of the tool is to be referenced as per the ACM0002 version 16. 3) IRR sheet is provided along with DPR and Loan Sanction Letter 5) As there are no other projects with the same technology and size within the CPA's parameters, Ndiff has been calculated as 0 in line with	
Documentation provided by project participant	
1) Updated VCS PD Version 1.2 dated 11 th January 2023	
VVB's assessment	Date: 20/07/2024
The additionality section and common practice analysis is applied and discussed as per the tools and VVB could verify the details provided in the PD with credible source. The additionality and common practice is justified and hence CAR is closed.	

CAR ID	08	Section no.		Date: 24/11/2023
Description of CAR				
1. The grid emission factor is required to determine as per Tool 7 (<u>Tool to calculate the emission factor for an electricity system</u>) and describe in PD the steps to calculate the CM value even though the reference document consider the steps. 2. The PD also transparently provide the source used for grid emission factor. 3. The monitoring plan should delineate the monitoring equipment's clearly showing the monitoring points.				
Project participant response				Date: 17/01/2024
As per the guidance of VCS V4.4, the PP submits the following 1) the default grid emission factor for Malawi has been sourced from the UNFCCC IFI Harmonized Default Grid Emission Factor v3.2 has been used as the latest publicly available dataset, the same is attached with this DVR for consideration 2) The same has been referenced within the PD and attached with this DVR 3) The monitoring plan specifies the project main meter and check meter and their serial numbers for monitoring of the project, the same is referred to within the PD.				
Documentation provided by project participant				
1) UNFCCC IFI Harmonized Default Grid Emission Factors v3.2 2) Updated VCS PD Version 1.2 dated 11th January 2023				
VVB's assessment				Date: 20/07/2024
The CM emission factor is sourced from UNFCCC (Harmonized IFI Default Grid Factors 2021 v3.2). The emission factor is published by International Financial Institutions technical working group on greenhouse gas accounting following 'methodological approach for the common default grid emission factor dataset'. The above methodology clearly states the approach of calculating CM following procedure stipulated in ACM0002 methodology considering Operating Margin (OM) and Build Margin (BM) and following AHSA-001 'IFI Approach to GHG accounting for renewable energy projects, version 2.0. Therefore, the CM emission factor calculated is as per the 'tool to calculate emission factor for electricity system' and correctly considered the value to be 0.3970 tCO₂e/MWh for Intermittent Energy (e.g., Solar, Wind, Tidal). The monitoring plan and arrangements are provided as per actual scenario and evident with monitoring procedure post commissioning. Hence, response is accepted and CAR is closed.				

APPENDIX C: COMPETENCY CERTIFICATES OF TEAM MEMBERS



Certificate of Appointment

<< CHAMPOK BURAGOHAIN >>

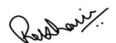
The qualification and appointment are done following the requirements of
ECOLANCE PRIVATE LIMITED (EPL/PR/5)

Standard/ Scheme			
☒ VCS	☒ CCBS	☒ GS4GG	☒ GCC
☐ SD Vista	☒ Social Carbon	☐ Plan Vivo	☐ Plastic Standard
☐ CerCarbono	☒ ISO 14064-1: 2018		

Role			
☒ Auditor	☒ Team Leader	☐ Reviewer	☒ Technical Expert
☐ Financial Expert	☐ Trainee	☐ Local Expert	☒ GHG Verifier [Applicable to 14064-1: 2018]

Sectoral Scopes	
☒ Energy Industries (Renewable/Non-Renewable)	
☒ Energy Demand	
☒ Waste Handling and Disposal	
☐ Forestry and Other Land Use	
☐ Agriculture	
☐ Livestock and Manure	
☒ General [Applicable to 14064-1: 2018]	

Certificate No	EPL/CER/CH/4
Date	15/02/2024
Validity	14/02/2025

Authorized by	Rekha Menon
Designation	Director
	



Certificate of Appointment

<< PREM KUMAR POKHREL >>

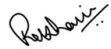
The qualification and appointment are done following the requirements of
ECOLANCE PRIVATE LIMITED (EPL/PR/5)

Standard/ Scheme			
☒ VCS	☐ CCBS	☒ GS4GG	☒ GCC
☐ SD Vista	☒ Social Carbon	☐ Plan Vivo	☐ Plastic Standard
☐ CerCarbono	☐ ISO 14064-1: 2018		

Role			
☒ Auditor	☒ Team Leader	☐ Reviewer	☒ Technical Expert
☐ Financial Expert	☐ Trainee	☒ Local Expert -Nepal	☒ GHG Verifier [Applicable to 14064-1: 2018]

Sectoral Scopes
☒ Energy Industries (Renewable/Non-Renewable)
☒ Energy Demand
☒ Waste Handling and Disposal
☐ Forestry and Other Land Use
☐ Agriculture
☐ Livestock and Manure
☐ General [Applicable to 14064-1: 2018]

Certificate No	EPL/CER/PP/3
Date	15/02/2024
Validity	14/02/2025

Authorized by	Rekha Menon
Designation	Director
	



Certificate of Appointment

<< ESWAR MURTY >>

The qualification and appointment are done following the requirements of
ECOLANCE PRIVATE LIMITED (EPL/PR/5)

Standard/ Scheme			
☒ VCS	☒ CCBS	☒ GS4GG	☒ GCC
☐ SD Vista	☒ Social Carbon	☐ Plan Vivo	☐ Plastic Standard
☐ CerCarbono	☒ ISO 14064-1: 2018		

Role			
☒ Auditor	☒ Team Leader	☒ Reviewer	☒ Technical Expert
☐ Financial Expert	☐ Trainee	☐ Local Expert	☒ GHG Verifier [Applicable to 14064-1: 2018]

Sectoral Scopes
☒ Energy Industries (Renewable/Non-Renewable)
☒ Energy Demand
☒ Waste Handling and Disposal
☐ Forestry and Other Land Use
☐ Agriculture
☐ Livestock and Manure
☒ General [Applicable to 14064-1: 2018]

Certificate No	EPL/CER/EM/1
Date	15/02/2024
Validity	14/02/2025

Authorized by	Rekha Menon
Designation	Director
